
IC 114: Linear Algebra

Course Code	IC 114
Course Name	Linear Algebra
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	B.Tech 1st year
Mutual Exclusion	None
Approval	50th BoA (Jul 2023); 42nd Senate (Jan 2024), in the IMBA (Integrated MBA) curriculum

Course Contents

- **Matrix Theory:** Rank of Matrix, inverse of a matrix by elementary operations, Solution of linear simultaneous equations and their numerical solutions by gauss Elimination and Gauss Seidel Methods. Eigen values and eigen vectors, Cayley Hamilton Theorem, Diagonalization of Matrices. Orthogonal, Hermitian, Skew Hermitian, Normal and Unitary matrices and their elementary properties, Quadratic Forms. [7 Lectures]
- **Vector Spaces:** Vector spaces, Sub Spaces, Linear Dependences and Independences of Vectors, Span, Bases and Dimensions, Direct Sum. [7 Lectures]
- **Linear Transformations:** Linear Transformations, Linear Variety, Range Space and Rank, Null Space and Nullity, Homomorphism, Matrix of Linear Transformations, Matrix Representation of a linear transformation, Structure of the solutions of the matrix equation $Ax = b$, Change of bases. [7 Lectures]

Text Books

1. G. Strang, **Introduction to linear algebra**, 4th Edition, Wellesley Cambridge Press.
2. Kenneth Hoffman and Ray Kunze, **Linear Algebra**, PHI publication.

References

1. NA